

1) What is the d/b var, let and const?

Feature	var	let	const
Scope	function scope	Block scope	Block scope
Re-declare	yes	No	No
Re-assign	yes	yes	No
Hoisting	yes	yes	yes

Example:

```
var a = 10;
```

```
let b = 20
```

```
const c = 30;
```

var → old method, function scoped

let → block scoped, value can change

const → block scoped, value can change

2) What are truthy and falsy values?

Falsy values:

Values that become false in condition checking.

false

0

""

null

undefined

NaN

Truthy Values:

All other values are truthy.

"hello"

1

[]

{}?

true

3) What is Hoisting in JavaScript?

Hoisting means JavaScript moves variable and function declarations to the top before execution.

Example:

```
console.log(a);
```

```
var a = 10;
```

Internally:

```
var a;
```

```
console.log(a);
```

```
a = 10;
```

var → hoisted with undefined
let and const → hoisted but cannot access before declaration

4) Difference between == and === ?

Operator

Meaning

==

checks value only

===

checks value and datatype

Example :

5 == "5" // true

5 === "5" // false

5) What is callback function?

A callback function is a function passed as an argument to another function.

~~Some of~~ the methods

- * function declaration
- * Anonymous function
- * Arrow function.

6) What is the difference between map() and forEach()?

map() :

- * Returns new array
- * Used for transformation
- * Can chain methods

Example:

```
const nums = [1, 2, 3];
```

```
const result = nums.map(n => n * 2);
```

```
console.log(result);
```

~~forEach()~~

- * Does not return array

- * Used for looping

- * Cannot chain easily

7) What is an arrow function? Difference from normal function?

Arrow function:

- Introduced in ES6 features

- work only inside the block.

Syntax:

```
const declaration = function () {  
  ;  
};
```

Normal function:

A normal function is defined using the function keyword and can have its own this, arguments and prototype.

Syntax:

```
let x = function function-name (parameters) {
```

```
  // body of the function  
  ;  
};
```


8) What is React? Why do we use it?

- * React is a JavaScript library used to build user interfaces.

- + It is efficient method.

- * Also created an single page application

- + It is an component based architecture.

Why use React?

- * Reusable components

- * Fast rendering using Virtual DOM

- * Easy state management

- * Single Page Application (SPA)

```
function App() {
```

```
  return <h> Hello </h>;
```

```
}
```

9) What is JSX? Can browsers understand JSX directly?

JSX means JavaScript XML.

It allows writing HTML inside JavaScript.

Example:

```
const element = <h1> Hello </h1>;
```

Example:

```
function greet (name) {  
  console.log ('Hello', + name + '!');  
}  
greet('Greet');
```

10) What is the difference b/w props and state?

Props	State
* Passed from parent * Read-only * Used for communication	* Managed inside component * Can change * Used for dynamic data

For example:

```
function welcome (props) {  
  return <h1> { props.name } </h1>;  
}
```

11) What is useState hook?
useState is used to create and update state in functional components.

Example:

```
import { useState } from 'react';
```

```
function counter () {
```

```
  const [count, setCount] = useState(0);
```

```
  return (
```

```
    <button onClick = { () => setCount(count + 1) } >
```

```
      {count}
```

```
    </button>
```

```
  );
```

```
}
```

12) What is useEffect hook? what does it run?
useEffect is used for side effects like:

- * API calls

- * timers

- * DOM updates

Example:

```
useEffect ( () => {
```

```
  console.log ( "Component Rendered" );
```

```
}, [] );
```

when it runs:

Dependency

Runs

no dependency → Every render

[]

→ only once

[value]

13) Virtual DOM:

Virtual DOM is a lightweight copy of the real DOM.

React compares changes in Virtual DOM and updates only changed parts in real DOM.

Advantages:

- * Faster updates
- * Better performance.

14) What happens when state updates in React? When state updates:

- 1) React re-renders the component.
- 2) Virtual DOM compares old and new changes
- 3) Only changed elements update in real DOM

Example:

```
setCount(count + 1);
```

This triggers re-rendering.

15) Why do we use key in .map() while rendering lists?

Keys help react identify which item changed, added or removed.

Example:

```
const users = [ "A", "B", "K" ];
```

```
users.map((user, index) => (
```

```
  <li key = { index } > { user } </li>  
  ));
```

Benefits:

- * Improves performance
- * Prevents unnecessary re-rendering
- * Helps React track elements correctly.